3D Scanner (Electrical)

Component selection:

- Stepper motor (nema 17 is more than enough) x2
- Stepper motor driver (A4988 or DRV8825) x2
- Start & stop buttons
- LCD 16x2: for user friendly experince (Giving user's instructions)
- SD card & SD card module: to save the scanned data and then transfer into a STL file
- Laser sensor: to measure the Depth and distance of the Present Part

Sensor Selection criteria:

- Range
- Resolution & Accuracy
- Compatability with other components & Cost
- Nature of output
- Sensitivity & Response time

Potenial Sensors:

TOF400C VL53L1X 4 Meter Laser Distance Sensor Module

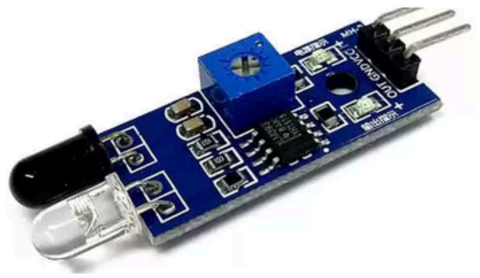
- Range: 4cm 400cm Maximum
- Dead zone: 0-4cm
- Wavelength: 940nm
- Viewing angle: 27 degrees
- Communication: I2C
- Minimum Resolution: 1 mm
- Typical Measurement Time: 30 ms – 100 ms
- Cost = 385 EGP

CNY-70 IR sensor

- Peak operating distance: < 0.5 mm
- Emitter wavelength: 950 nm
- Include Daylight blocking filter
- Only detects black/white surfaces
- Cost = 25 EGP

Ultrasonic Sensor Module HY-SRF05

- Range = 2cm -450cm
- Resolution = 2mm
- 40 kHz operation
- ±15 degree field of view
- Communication: UART
- Cost = 80 EGP



Infrared IR Obstacle Avoidance Sensor

- Range = 2 – 30cm
- 35 degrees field of view
- Cost = 30 EGP
- The output is in the form: DO digital switching outputs (0 and 1) and AO analog voltage output.
- Only detect presence (not depth)

Conclusion:

- The TOF400C VL53L1X and Ultrasonic sensors are the best fit, However each has it's own limitations.
  - Limitations of an Ultrasonic Sensor:
    - Low accuracy (~3mm to 10mm error) → Less precise than a laser sensor.
    - Wide beam angle (~15° to 30°) → Can't detect small details.
    - Reflective issues → Hard surfaces reflect sound in different directions, causing errors.
  - Limitations of TOF400C VL53L1X:
    - Short-range (~1m): High accuracy (~±5mm).
    - Long-range (~4m): Accuracy decreases (~±50mm or more).
    - Affected by surface reflectivity, ambient lighting, and angle of the object.
    - Faster Measurements (~20ms per reading) → Less accuracy (more noise).
    - Slower Measurements (~100ms per reading) → Higher accuracy (less noise).